

CMSC 6940

Computer Based Research Tools and Applications

Course Details

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*Class: ED1005, Education Building
Held alternate Fridays starting on September 17, 2010
Time: 3:30-5:50 pm
Lab Room: C3057/61, Chemistry-Physics Building
Held alternate Fridays starting on September 24, 2010
Time: 3-5 pm*

Course Objectives

The objective of the course is to introduce students to computational tools that they will use in their research to complement codes that they will develop. This course provides an introduction to the role and application of several computational tools in scientific research. The course will be partitioned into modules designed to cover specific topics such as the application of specific packages for mathematical modeling, task automation, data representation and analysis.

Before beginning this course, students should have:

- a) a basic understanding of the linux operating system and environment
- b) some basic understanding of algorithms
- c) a reasonable understanding of their research project, sufficient to apply related applications to their research area

Assigned Work

Assignments – 4 @ 10% each
Final Exam – 60%

Each assignment is designed to help the student apply what has been learned in the modules to their research. As the class will run every two weeks, in the alternate weeks there will be a lab to help the student practice what was learned in the modules and to work on the assignments. Extra exercises will be provided and will be geared towards building experience with the tools covered in the modules as applied to different research areas. The final exam is meant to tie together the different tools outlined in the modules. Further details on the exam format will be given later in the course.

Course Contents

The course will consist of 7 classes of 2 hours each, with each class being a module. Students will be tested, through the assignments and the exam, on the contents of each of these modules.

Shell Scripting

- Introduction to tcsh
- Introduction to bash

Introduction to Perl

- Overview
- Variables, Loops
- Modules, Functions
- Useful Add-ons

Introduction to Python

- Overview
- Python Parts, Relating to Perl
- Introduction to NumPy and SciPy

Software Building Tools

- Intro to Autotools
- Intro to Make
- Compilers, Preprocessors

Profiling and Debugging

- Introduction to gdb and ddd
- Quick Overview of TotalView for Debugging
- Introduction to Strace and Valgrind for Profiling

Mathematical Scripting

- Introduction to Octave
- Other Math Scripting Tools

Visualization

- Basics of OpenDX
- Some Tools Specific to Research Areas

Required Text

There will be no text for this course and students will be expected to do their own searches for the course material. Many of the topics discussed here have excellent online resources. However, some reference materials are listed below as a good starting point.

Additional Reading

Programming Perl, by Larry Wall et. al., O'Reilly Media, 2003

Perl Cookbook, by T. Christianson and N. Torkington, O'Reilly Media, 2003

Python Scripting for Computational Science, by H. P. Langtangen, Springer, 2007

Learning Python, by Mark Lutz, O'Reilly Media, 2009

Python Cookbook, by A. Martelli et. al., O'Reilly Media, 2005

A Practical Guide to Linux Commands, Editors and Shell Programming, by Mark Sobell, Prentice Hall, 2010

The Linux Programmer's Toolbox, by John Fusco, Prentice Hall, 2007

GNU/Linux Application Programming, by M. Tim Jones, Charles River Media, 2008

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